# Hierarchical Bayesian Modeling of *In-Play* NFL Win Probability

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**Abstract**

Predicting win probability (WP) in real-time during National Football League (NFL) games is challenging due to dynamic game states and varying team strengths. This paper develops and compares two Bayesian Hierarchical Models (BHMs) for estimating in-play NFL win probability using play-by-play data from the 2022-2024 seasons (nflverse). The base model uses a fixed home-field advantage, while the extended model incorporates team-specific random effects for both baseline strength and home advantage, explicitly modeling opponent strength. Using `NumPyro` and Hamiltonian Monte Carlo (NUTS), we find comparable predictive accuracy (AUC $\approx$ 0.816 vs. 0.818 for Base vs. Extended). Posterior analysis highlights score differential's importance and reveals significant heterogeneity in team strength and home advantage captured by the extended model. Scenario analyses confirm model responsiveness, and an exploratory analysis suggests a modest, positive correlation (Kendall's $\tau \approx 0.217$) between estimated home advantage and fan attendance rankings. This BHM framework offers a flexible, interpretable approach for real-time NFL WP, providing inherent uncertainty quantification.

## 1 Introduction

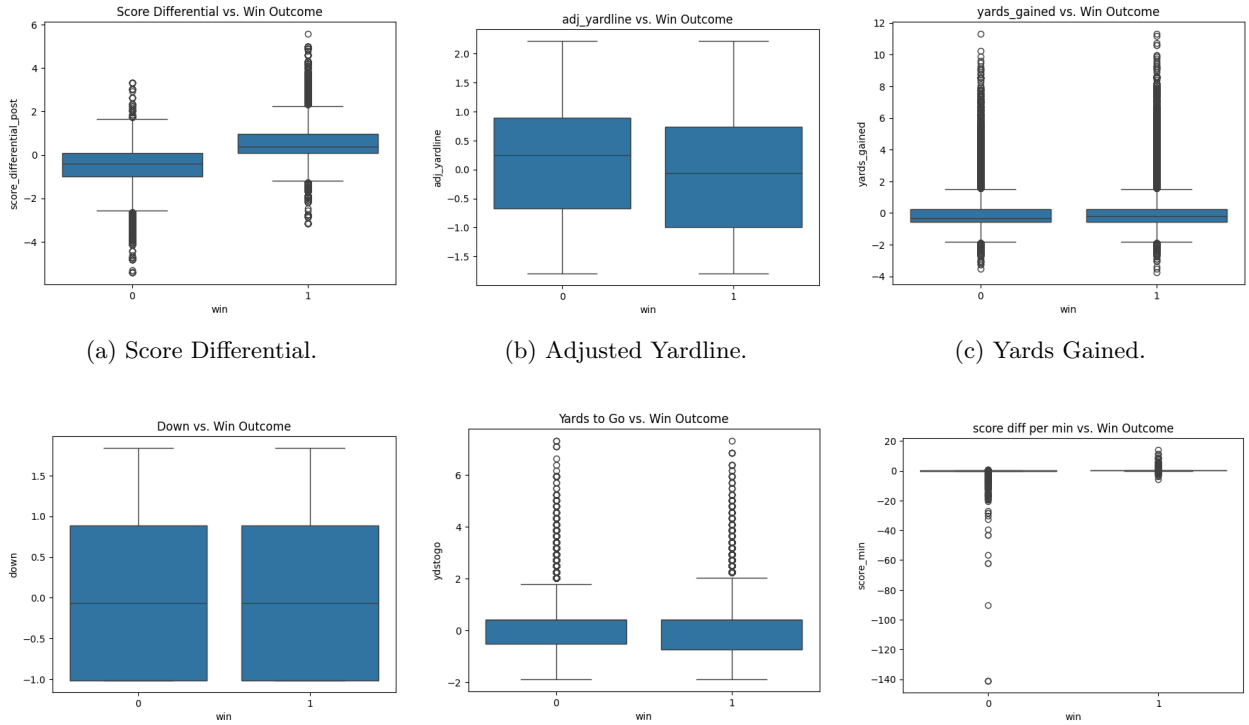
Estimating real-time win probability (WP) in the National Football League (NFL) is complex, given dynamic game situations and latent team factors. While traditional methods like simple logistic regression exist, Bayesian Hierarchical Models (BHMs) offer advantages by incorporating hierarchical structures (e.g., team-level effects within league trends) and naturally quantifying prediction uncertainty (Gelman et al., 2013). This study applies BHMs to model in-play NFL WP using modern Bayesian inference techniques. We develop and compare two specific models: a base model with a fixed home-field advantage (HFA) and team strength random effects, and an extended model incorporating team-specific random effects for HFA and explicitly modeling opponent strength. Our primary goal is to provide a flexible probabilistic framework that disentangles team strength from HFA, quantifies uncertainty, and explores potential drivers of HFA, contributing to real-time sports analytics.

## 2 Data Acquisition and Pre-processing

Play-by-play data for the 2022-2024 NFL regular seasons were obtained from the nflverse-data repository. We selected relevant game-state, scoring, and team variables. Filtering retained core play types (pass, run, kicks) and any plays resulting in a score change. Missing values for down/distance on extra points were imputed, and records with other missing essential predictors were removed. Key engineered features include the binary target variable 'win' (1 if possessing team won), 'minutes_remaining', 'adj_yardline' (field position adjusted for yards gained), a binary 'is_home' indicator, and 'score_min' (clipped score differential per minute). Kickoffs and no-plays were excluded. The data was split 80/20 into training/testing sets based on game ID to prevent leakage. Continuous predictors were standardized using a scaler fit only on the training data. Team indices were created based on the training set for use in the hierarchical models. The complete workflow, including data processing, exploratory analysis, model specification in `NumPyro`, and result generation, can be reviewed in the accompanying Google Colaboratory notebook, accessible at: `https://colab.research.google.com/drive/1SDqoVNLuZScxA5-mhNP8L2kyrJ9m-VXn?usp=share_link`.

# 3 Data Exploration

Exploratory data analysis involved visualizing the relationship between key predictor variables and the win outcome using boxplots. As seen in Figures 1a through 1f, moderation associations were confirmed for variables directly reflecting game advantage, such as post-play score differential (Figure 1a), adjusted yardline (Figure 1b), and score differential per minute (Figure 1f). Variables like down, distance to go, and yards gained on a specific play showed unbiased relationships with the final win outcome (Figures 1c, 1d, 1e).




(a) Score Differential. (b) Adjusted Yardline. (c) Yards Gained.

(d) Down. (e) Yards to Go. (f) Score Diff./Min.

Figure 1: Boxplots comparing predictor variable distributions vs. Win Outcome.

# 4 Prior Literature

Modeling win probability (WP) in sports has advanced alongside data availability. While early approaches used simpler regressions, the advent of detailed play-by-play data, accessible via sources like nflverse (Tan Ho et al., 2022), has enabled sophisticated real-time models. Frequentist methods, including machine learning techniques, are common for public WP models but often lack inherent uncertainty quantification (Lock & Nettleton, 2014). Bayesian Hierarchical Models (BHMs) address this by providing full posterior distributions and allowing for principled modeling of group-level effects (like team strength) through partial pooling (Baio & Blangiardo, 2010). Home-field advantage (HFA) is a recognized factor (Courneya & Carron, 1992), and while sometimes treated as fixed, evidence suggests it can vary by team, supporting the random effects approach used in our Extended Model. Our work applies BHMs specifically to in-play NFL WP, comparing fixed vs. random HFA effects and exploring correlates like fan attendance, contributing to the literature on dynamic sports modeling within a probabilistic framework (Gelman et al., 2013).

# 5 Model Specification

We constructed two Bayesian hierarchical models to estimate the win probability in NFL games.

1. **Base Model**: This model assumes that all 32 NFL teams share a common home advantage, represented as a fixed effect.
2. **Extended Model**: This model allows for team-specific home advantages by introducing a random effect for the home advantage, capturing variation across teams.

## 5.1 Base Model - Fixed Home Advantage

For play $i$ with possession team $j$:

$$
\begin{aligned}
y_{i,j} &\sim \text{Ber}(p_{i,j}) \\
\text{logit}(p_{i,j}) &= \underbrace{x_{i,j}^T\beta}_{\text{Fixed effect 1}} + \underbrace{z_{i,j}^T\gamma}_{\text{Fixed effect 2}} + \underbrace{\alpha_j}_{\text{Random effect}}
\end{aligned}
$$

- $y_{i,j}$ is the binary outcome (win=1, loss=0) for possession team $j$ in observation $i$.
- $x_{i,j}$ represents the game situation covariates (time remaining, score differential, adjusted yardline, etc.)
- $z_{i,j}$ is a home team dummy variable.
- $\gamma$ represents the home advantage effect, which is shared across all teams.
- $\alpha_j$ is the team-specific strength parameter for team $j$ in play $i$.

We assumed that the team-specific random effects $\alpha_j$ follow a normal distribution with mean $\mu_{team}$ and variance $\sigma^2_{team}$. For Bayesian estimation, we assigned weakly informative priors to the parameters $\beta$, $\gamma$, $\mu_{team}$, and $\sigma^2_{team}$ as specified below. Although prior distributions can incorporate domain knowledge when available, we chose weakly informative priors to ensure that the posterior distributions are primarily driven by the observed game data rather than by prior assumptions.

$$
\begin{aligned}
\alpha_j \sim & N(\mu_{team}, \sigma^2_{team}) \\
\beta \sim N(0, 2^2 I), \quad & \gamma \sim N(0, 2^2 I) \\
\mu_{team} \sim N(0, 1), \quad & \sigma^2_{team} \sim HalfNormal(1)
\end{aligned}
$$

## 5.2 Extended Model - Team-Specific Home Advantage

Unlike the base model, which assumes that all teams share a common home advantage effect, the extended model allows for variability in home advantage across teams by modeling team-specific home advantages as random effects. Additionally, to capture the dynamics of individual matchups, we included the difference between the team-specific strength parameters of the possession team and the opposing team in the model. The game situation covariates $X_{i,j}$ were treated as fixed effects, consistent with the base model.

$$
\begin{aligned}
y_{i,j} &\sim \text{Ber}(p_{i,j}) \\
\text{logit}(p_{i,j}) &= \underbrace{x_{i,j}^T\beta}_{\text{Fixed effect 1}} + \underbrace{\alpha_j - \alpha_k + \delta_j \cdot I(home_i)}_{\text{Random effect}}
\end{aligned}
$$

- $\alpha_j$ is the team-specific strength parameter for possession team $j$.
- $\alpha_k$ is the team-specific strength parameter for opponent team $k$.
- $\delta_j$ represents the team-specific home advantage parameter for team $j$.

- $I(home_i)$ is an indicator function that equals 1 when the possession team is the home team in play $i$, and 0 otherwise.

As in the base model, the team-specific strength parameters $\alpha_j$ and the home advantage parameters $\delta_j$ were assumed to follow normal distributions. Additionally, weakly informative priors were assigned to all parameters required for posterior inference, consistent with the previous specification.

$$\begin{aligned} \alpha_j \sim N(\mu_{team}, \sigma^2_{team}), \quad & \delta_j \sim N(\mu_{home}, \sigma^2_{home}) \\ \mu_{team} \sim N(0,1), \quad & \sigma^2_{team} \sim HalfNormal(1) \\ \mu_{home} \sim N(0,1), \quad & \sigma^2_{home} \sim HalfNormal(1) \end{aligned}$$

In both the base model and the extended model, we applied a non-centered parameterization technique for sampling the parameters $\alpha$ and $\delta$. When $\alpha$ and $\delta$ are sampled directly from their respecitve normal distributions, small values of $\sigma_{team}$ or $\sigma_{home}$ can lead to strong correlations between $\alpha$ and $\delta$, and samples of $\mu$ and $\sigma$. These correlations hinder the MCMC sampler's ability to efficiently explore the parameter space, resulting in poor mixing and slow convergence.

Non-centered parameterization mitigates this issue by sampling auxiliary variables $\alpha_j^{raw}$ and $\delta_j^{raw}$ instead of $\alpha_j$ and $\delta_j$ directly, and then computing:

$$\alpha_j = \mu_{team} + \sigma_{team} \cdot \alpha_j^{raw}, \quad \delta_j = \mu_{home} + \sigma_{home} \cdot \delta_j^{raw}$$

$$\alpha_j^{raw},\ \delta_j^{raw} \sim N(0,1)$$

This reparameterization reduces the correlations among posterior samples, thereby improving the efficiency of the MCMC sampler. This technique is particularly effective within the Hamiltonian Monte Carlo framework, which we employed for estimation.

# 6 Inference Method

Our hierarchical model involves estimating a large number of parameters, including team-specific strength parameters, team-specific home advantage parameters, and their associated means and variances. In the base model, we estimate a total of 42 parameters: 7 coefficients for the fixed effect variables related to game situations, 1 common home advantage coefficient, 32 team-specific strength parameters, and 2 hyperparameters. In the extended model, the inclusion of team-specific home advantage effects increases the number of parameters to over 70, all of which require posterior distribution estimation. While Gibbs sampling and the Metropolis-Hastings algorithm are commonly used for Bayesian model estimation, they may not be suitable for models with such a high number of parameters. Both methods are known to suffer from poor mixing and slow convergence when applied to high-dimensional posterior distributions, making them inefficient for this context.

## 6.1 Hamiltonian Monte Carlo

As an alternative, we employed the Hamiltonian Monte Carlo (HMC) method, specifically the No-U-Turn Sampler (NUTS), to estimate our model. HMC, inspired by principles from physics, enhances sampling efficiency by utilizing the gradient of the log-posterior function to guide exploration of the parameter space. Also, by introducing auxiliary momentum variables, HMC enables more informed proposals, allowing the sampler to traverse the parameter space more effectively compared to traditional MCMC methods. For a target density $\pi(x)$, where $x \in \mathbb{R}^d$, and an auxiliary momentum variable $z \in \mathbb{R}^d$, the joint distribution of $x$ and $z$ can be expressed using the Hamiltonian formulation as follows:

$$p(x,z) \propto \exp\left(-H(x,z)\right),$$

where $H(x,z) = -\log \pi(x) + \frac{1}{2} z^T M^{-1} z$ denotes the Hamiltonian, and $M$ is a positive definite mass matrix.

The Hamiltonian can be interpreted as the sum of potential energy and kinetic energy, and, according to the principle of energy conservation, $H$ remains constant as the dynamical system evolves over time. This property allows HMC to efficiently explore regions of high probability in the target distribution. To describe Hamiltonian dynamics in discrete time, the dynamical system can be expressed as follows:

$$x_{t+1} = x_t + \epsilon\, \nabla_z H(x_t, z_t) = x_t + \epsilon\, M^{-1} z,$$
$$z_{t+1} = z_t - \epsilon\, \nabla_x H(x_t, z_t) = z_t - \epsilon\, \frac{\partial}{\partial x_i} log\pi(x),$$

where $\epsilon$ is the step size, and $\nabla_x H$, $\nabla_z H$ denote the gradients of the Hamiltonian with respect to $x$ and $z$, respectively.

If $\pi(x)$ is flat, then $\nabla \log \pi(x) = 0$, and $x$ moves at a constant velocity. However, if $\pi(x)$ is not flat, $x$ is accelerated by $\nabla \log \pi(x)$ and moves towards regions of higher density. Using this formulation, the Hamiltonian Monte Carlo (HMC) algorithm can be described as follows:

1. Sample $z \sim \mathcal{N}(0, M)$ where $M = \text{diag}(m_1, \ldots, m_d)$.
2. Set $x_0 \leftarrow x$ and $z_0 \leftarrow z$.
3. For $\ell = 1, \ldots, L$,
   (a) $z \leftarrow z + \frac{1}{2}\epsilon \nabla \log \pi(x)$
   (b) $x \leftarrow x + \epsilon M^{-1} z$
   (c) $z \leftarrow z + \frac{1}{2}\epsilon \nabla \log \pi(x)$
4. Metropolis accept/reject step:
   (a) $\alpha \leftarrow \min\left\{1, \frac{\pi(x)\,\mathcal{N}(z|0,M)}{\pi(x_0)\,\mathcal{N}(z_0|0,M)}\right\}$
   (b) With probability $\alpha$, accept $x$ as the new state, otherwise reject and keep the state as $x_0$.

Step 3 of the algorithm (leapfrog integration) describes the discretization of the continuous dynamical system, and the proposed value of $x$ is accepted or rejected through the Metropolis accept/reject step in Step 4.

## 6.2 No-U-Turn Sampler (NUTS)

Although HMC significantly improves sampling efficiency, one of its main challenges is the manual tuning of the trajectory length, determined by the number of leapfrog steps $L$. If $L$ is too small, the sampler produces highly correlated samples and exhibits slow mixing. Conversely, if $L$ is too large, the sampler may backtrack, leading to unnecessary computations and inefficiency.

The No-U-Turn Sampler (NUTS) addresses this issue by automatically determining an appropriate trajectory length during sampling. NUTS adaptively stops the trajectory when it begins to reverse direction, thereby eliminating the need for manual selection of $L$. This adaptive mechanism prevents redundant exploration while preserving detailed balance, ensuring the correctness of the sampling process. Furthermore, NUTS facilitates more efficient exploration of the parameter space, especially in high-dimensional settings, and is readily implemented in probabilistic programming frameworks such as `NumPyro`.

Specifically, NUTS adaptively determines the step size $\epsilon$ and the mass matrix $M$ during the warm-up period. Based on the values of $\epsilon$ and $M$ obtained during warm-up, NUTS dynamically adjusts the number of leapfrog steps $L$ at each iteration in the subsequent sampling phase.

To estimate our model, we employed `NumPyro` in a Google Colab environment utilizing an A100 GPU. A total of 1,000 samples were used for the warm-up phase, followed by an additional 1,000 samples collected for posterior inference.

### 6.3 Sampling Diagnostics

The efficiency and reliability of Markov Chain Monte Carlo (MCMC) sampling were carefully assessed through standard diagnostic measures. For both Model 1 and Model 2, that will be explained further in later chapters, convergence was evaluated using the potential scale reduction factor ($\hat{R}$), effective sample size (ESS), and divergence detection.

Across all parameters in both models, the $\hat{R}$ values were approximately 1.00, indicating that the chains had converged to the target posterior distribution without substantial between-chain variability. In addition, the effective sample sizes for all key parameters - including fixed effect coefficients ($\beta$), team effect parameters, and home advantage components - were sufficiently large, ensuring that posterior estimates were based on independent information rather than highly autocorrelated samples.

The average acceptance probability during the NUTS sampling phase was approximately 0.92, consistent with the recommended target range (0.8-0.95) for Hamiltonian Monte Carlo methods. This high acceptance rate implies that the sampler made efficient proposals, leading to good exploration of the posterior space while avoiding excessive rejection or retracing. Minimum ESS exceeding 80/100 across all parameters indicates sufficiently independent samples for reliable posterior estimation. Importantly, no divergences were recorded during the sampling of either model, further reinforcing that the sampling trajectories were well-behaved.

Table 1 summarizes the key diagnostic metrics for both models. Overall, these results demonstrate that the NUTS in `NumPyro` successfully generated high-quality posterior samples, enabling robust downstream inference and prediction.

Table 1: Sampling Diagnostics for Model 1 and Model 2

| Model | Average $\hat{R}$ | Minimum ESS (over all parameters) | Acceptance Rate | Divergences |
|---|---|---|---|---|
| Model 1: Fixed Home Adv | 1.00 | $> 80$ | 0.94 | 0 |
| Model 2: Varying Home Adv | 1.00 | $> 100$ | 0.94 | 0 |

# 7 Results

### 7.1 Posterior Summaries and Fixed Effects

To assess the learned relationships between game features and win probability, we compared posterior summaries from two models: one with a fixed home-field advantage shared across all teams (Model 1), and one with varying home-field advantage for each team (Model 2).

Table 2: Predictive performance on test set (mean $\pm$ 1 SD over posterior draws)

| Parameter | Description | Model 1 | Model 2 |
|---|---|---|---|
| $\beta_0$ (minutes_remaining) | Time remaining in the game | $-0.06 \pm 0.01$ | $-0.07 \pm 0.01$ |
| $\beta_1$ (score_differential) | Current score differential | $1.47 \pm 0.02$ | $1.49 \pm 0.02$ |
| $\beta_2$ (adj_yardline) | Distance to opponent's end zone | $-0.13 \pm 0.01$ | $-0.14 \pm 0.01$ |
| $\beta_3$ (yards_gained) | Yard gained on the play | $0.06 \pm 0.01$ | $0.06 \pm 0.01$ |
| $\beta_4$ (down) | Current down | $-0.14 \pm 0.01$ | $-0.13 \pm 0.01$ |
| $\beta_5$ (ydstogo) | Distance to first down | $-0.03 \pm 0.01$ | $-0.03 \pm 0.01$ |
| $\mu_t$ (Team strength) | Average team effect | $-0.20 \pm 0.10$ | $-0.01 \pm 1.04$ |
| $\sigma_t$ (Team variance) | Spread of team effect | $0.62 \pm 0.07$ | $0.68 \pm 0.09$ |
| $\gamma$ (Shared Home Adv) | Same home adv for all teams | $0.26 \pm 0.02$ | - |
| $\mu_h$ (Home Adv Mean) | Average team-specific home adv | - | $0.12 \pm 0.08$ |
| $\sigma_h$ (Home Adv Variance) | Spread in home adv across teams | - | $0.42 \pm 0.07$ |
| $\delta_i$ (Team Home Adv) | Team-specific home adv | - | 32 team-specific values |

According to Table 2, the posterior means for the fixed-effect coefficients ($\beta_0$ to $\beta_5$) are nearly identical across both models, as expected given the same underlying dataset and covariate structure. The coefficient for score differential ($\beta_1$) is the largest in magnitude, indicating that current game score remains the strongest predictor of win probability.

A notable distinction appears in the modeling of home-field advantage. Model 1 assumes a uniform home advantage, whereas Model 2 introduces latent team-specific deviations. The posterior spread of team-specific advantages ($\sigma_h \approx 0.41$) reveals substantial heterogeneity among teams in their benefit from playing at home.

In particular, Model 2 captures richer inter-team dynamics by allowing each team to have its own baseline strength (team effect) and unique home-field influence. The larger posterior variance for $\mu_t$ in Model 2 reflects the model's flexibility in learning these effects jointly with home advantage.

To further illustrate the uncertainty and directionality of these effects, we visualize the posterior distributions of the six covariates ($\beta_0$ to $\beta_5$) for both models. As shown in Figure 2 and Figure 3, the posterior distributions are consistently tight and clearly distinguishable from zero for most covariates - especially score differential ($\beta_1$), which appears to be the most influential.

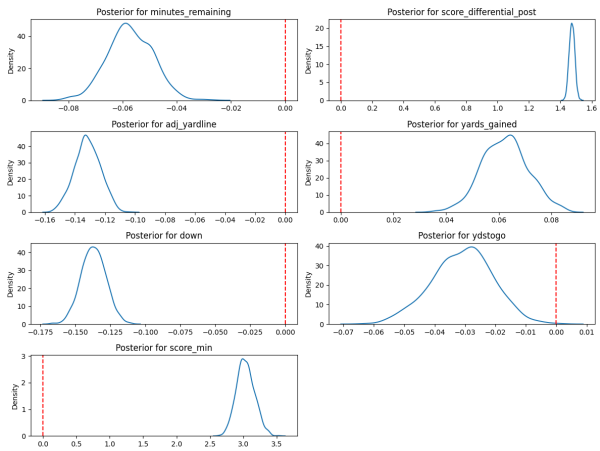



Figure 2: Posterior Distributions for Model 1

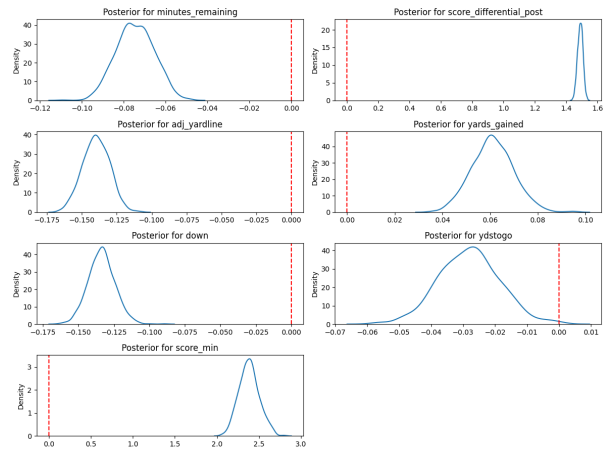



Figure 3: Posterior Distributions for Model 2

It can be noticed that there are 7 posteriors in the figures above, even though only 6 features were used. The last variable is score_differential / time_remaining. This helps the model to capture the non-linear relationship between those two variables, which was discovered while experimenting with various scenarios, which will be covered in a later part of this paper.

## 7.2 Team Effects

The model also includes latent team effects to account for unobserved heterogeneity in performance. Figure 4 presents the team effects from Model 1. Here, positive values correspond to teams that consistently outperform expectations, while negative values indicate underperformers. Teams such as Kansas City (KC), Philadelphia (PHI), and Detroit (DET) show positive effects.

Model 2 similarly incorporates team effects but allows for additional variance due to team-specific home advantage. Figure 5 shows that although the general raking of team strength remains consistent with Model 1, the confidence intervals are wider - reflecting increased uncertainty due to model complexity. This is because Model 2 includes both average team effect, and 32 individual team effect raw parameters. In this formulation, there is no unique way to allocate the overall mean team strength between $\mu_t$ and centered team effects. As a result, the model needs to allocate more variability to team strengths to jointly account for both neutral performance and latent contextual differences, which leads to higher variance in team effects.

## 7.3 Team-Specific Home Advantage

Unlike Model 1, Model 2 estimates a unique home advantage for each team. Figure 6 displays these effects. Teams such as Cleveland (CLE), Carolina (CAR), and Denver (DEN) are estimated to benefit

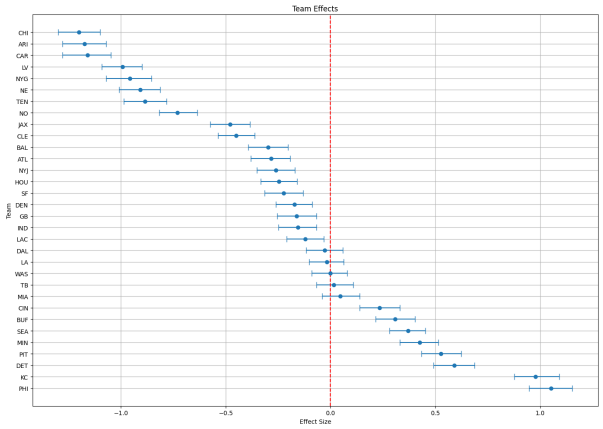



Figure 4: Team Effect Estimates (Model 1)

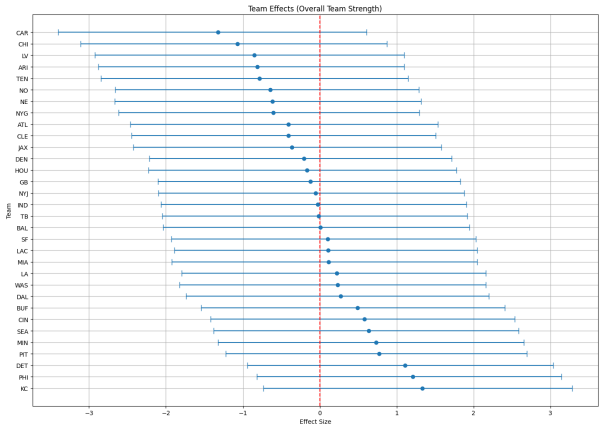



Figure 5: Team Effect Estimates (Model 2)

from a significantly stronger home field advantage, while teams like Detroit (DET), Indianapolis (IND), and Seattle (SEA) show relatively weaker or negligible effects.

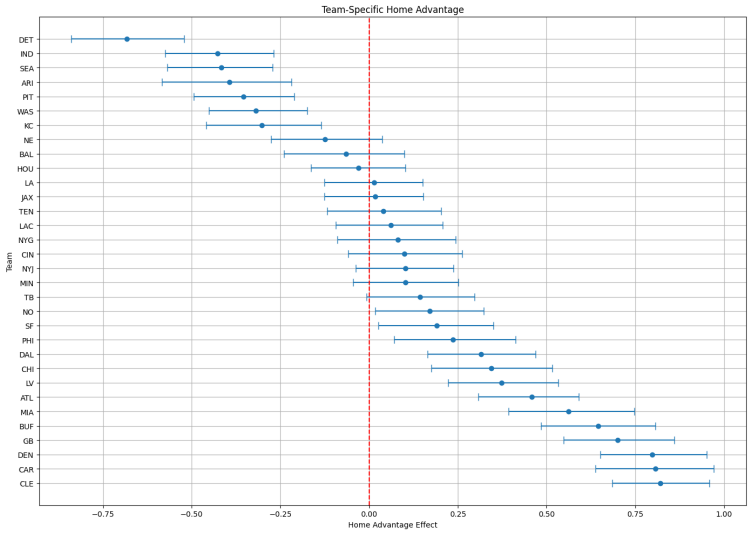



Figure 6: Team-Specfic Home Advantage Effects (Model 2)

## 7.4 Model Evaluation

To assess predictive performance, we evaluated both models on a held-out test set using four standard classification metrics: accuracy, Area Under the ROC Curve (AUC), Brier score, and log loss. Additionally, we report the confusion matrix components (true positives, false positives, true negatives, false negatives) for a comprehensive view of prediction outcomes.

Based on accuracy, both models perform nearly identically. Model 2 slightly outperforms Model 1 in terms of AUC, and this suggests that the added flexibility from team-specific home advantage helps in ranking predictions more effectively. Both models have similar Brier score and Log loss as well. The confusion matrices reveal that Model 2 captures more true positives and fewer false negatives, suggesting it may be more sensitive in identifying winning plays. These observations are further supported by the ROC curves shown in Figure 7 and Figure 8. The curves for both models demonstrate a consistent improvement over random guessing.

Table 3: Model Evaluation

| Metric | Model 1 | Model 2 |
|---|---|---|
| Accuracy | 0.7332 | 0.7313 |
| AUC | 0.8157 | 0.8177 |
| Brier Score | 0.1757 | 0.1761 |
| Log Loss | 0.5212 | 0.5221 |
| TP/FP | 8417/2842 | 8865/3333 |
| TN/FN | 8344/3258 | 7853/2810 |

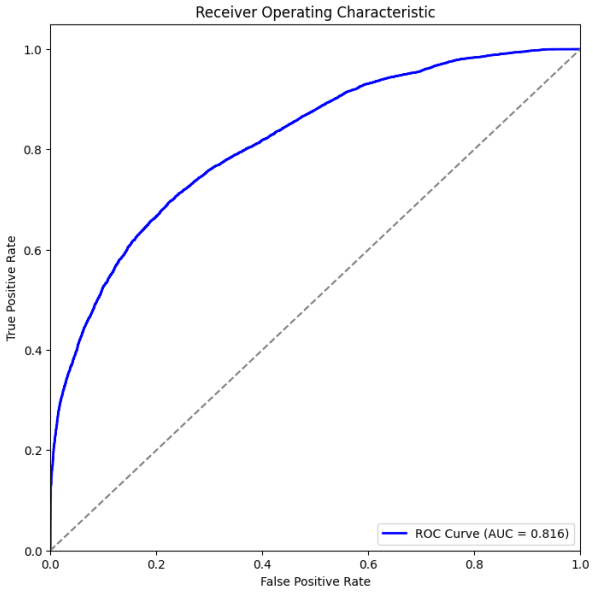



Figure 7: ROC Curve (Model 1)

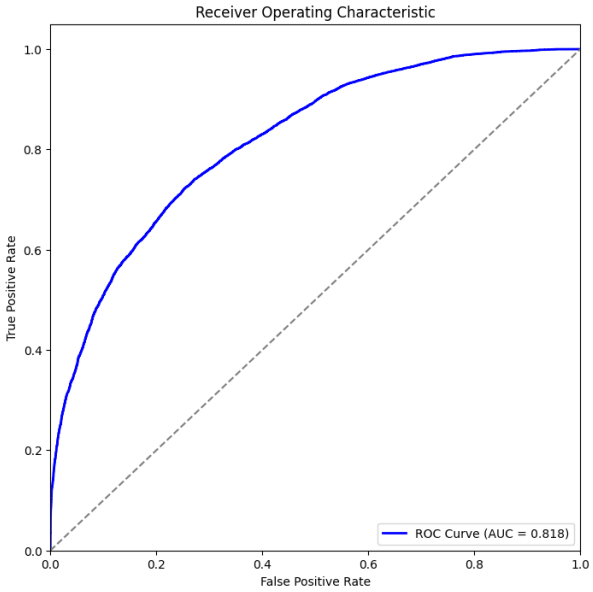



Figure 8: ROC Curve (Model 2)

From a Bayesian modeling perspective, Model 2 also offers greater flexibility and interpretability by capturing team-specific home advantages. This added granularity, even if slightly more complex, enables richer downstream insights, as discussed in the next section.

# 8 Scenario Analysis

To complement our model evaluation and posterior interpretation, we perform scenario-based analysis to understand how the model responds to different in-game contexts. Three representative scenarios were examined. These scenarios were selected to represent a range of in-game situations: a moderately challenging context, an extremely unfavorable context, and a highly favorable context. Each scenario was input into both the fixed and varying home advantage models to analyze predict win probabilities across all NFL teams.

## 8.1 Scenario 1: Moderate Situation

Game state:

- 10 minutes remaining
- 6 points behind
- Ball at own 30-yard line (adj_yardline = 70)

- 0 yards gained on the play
- 2nd down
- 10 yards to go

This scenario represents a common mid-game situation where the offensive team has a chance to shift momentum despite being behind.

Figure 9 shows the top 10 teams with the highest win probabilities in this scenario when they are the home team. The range of predicted win probabilities spans from approximately 0.46 to 0.7. Teams such as Philadelphia (PHI) and Buffalo (BUF) rank highest, with probabilities close to 0.7, signaling a substantial improvement in likelihood of victory under home conditions.

Figure 10 presents the same scenario with teams as the away team. The distribution shifts leftward for most teams, ranging from approximately 0.42 to 0.68. This reflects a certain amount of hinder on team performance due to lack of the home-field advantage. Kansas City (KC), Philadelphia (PHI), and Detroit (DET) rank highest, with probabilities close to 0.67, indicating stronger resilience in hostile environments.

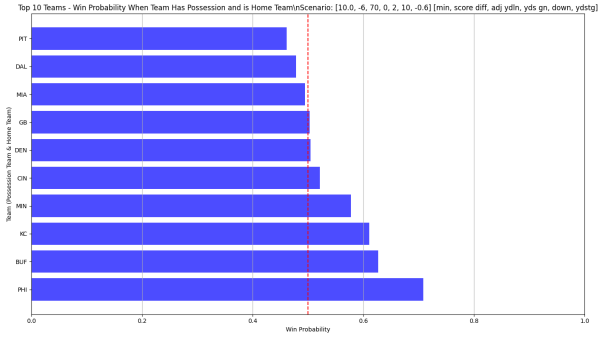



Figure 9: Top 10 teams by predicted win probability at home (Scenario 1)

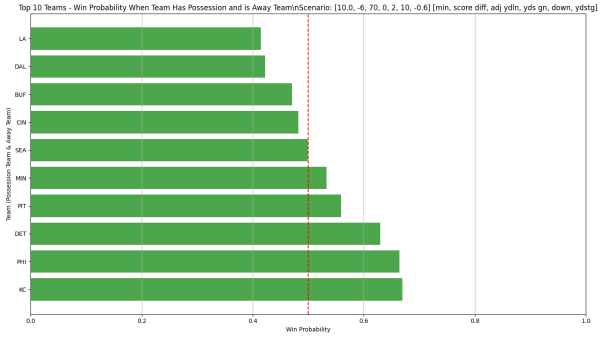



Figure 10: Top 10 teams by predicted win probability at away (Scenario 1)

To further dissect the contribution of team-specific home-field advantage, Figure 11 quantifies the marginal effect of playing at home on predicted win probability. The effect is computed as the difference between home and away probabilities for each team. Denver (DEN), Cleveland (CLE), and Green Bay (GB) show the most pronounced home advantage effects, improving win probability by over 0.17 under home conditions. This aligns with the model's posterior distribution of team-specific home advantage learned during training.

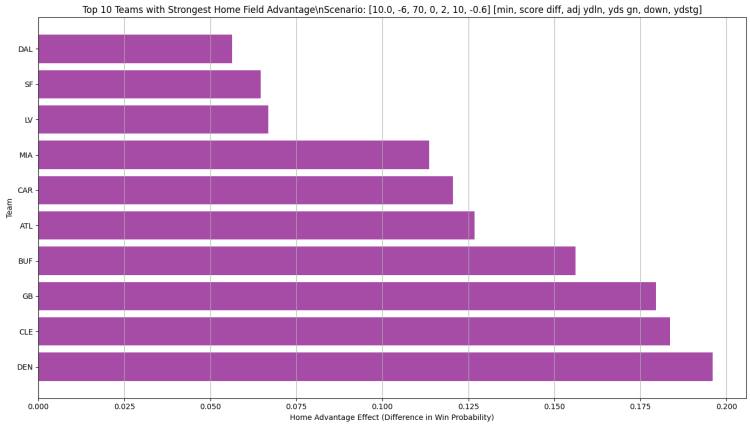



Figure 11: Top 10 teams by estimated home-field advantage effect (Scenario 1)

## 8.2 Scenario 2: Unfavorable Situation

Game state:

- 4 minutes remaining
- 8 points behind
- Ball at own 10-yard line (adj_yardline = 90)
- 0 yards gained on the play
- 4th down
- 25 yards to go

The scenario presented represents a highly adverse game situation, likely leading to a turnover or failed conversion attempt. According to Figure 12, it is evident that win probabilities decline across the board. Even the most capable teams under the model, such as Philadelphia (PHI), Buffalo (BUF), and Kansas City (KC) not being able to reach 0.4. This reflects how difficult it is to mount a comeback in this context, even when they are at home.

When away, Figure 13 shows that the model assigns even lower win probabilities. For most teams, the prediction hovers around or below 0.30.

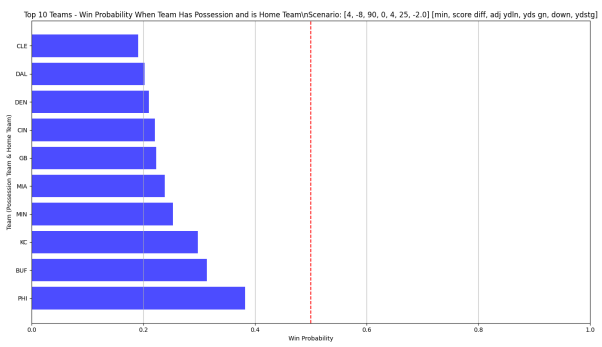



Figure 12: Top 10 teams by predicted win probability at home (Scenario 2)

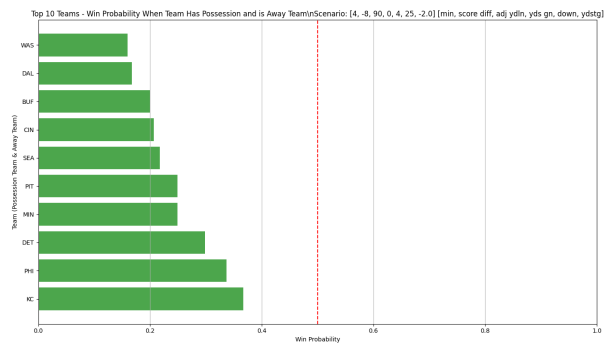



Figure 13: Top 10 teams by predicted win probability at away (Scenario 2)

It can be seen that home advantage for teams are lower compared to Scenario 1 as well, with the highest being 0.11, while it was almost 0.2 in Scenario 1. This is because being at home or away will have little influence in such a desperate scenario.

## 8.3 Scenario 3: Favorable Situation

Game state:

- 2 minutes remaining
- 5 points behind
- Ball at the opponent's 10-yard line (adj_yardline = 10)
- 5 yards gained on the play
- 1st down
- 5 yards to go

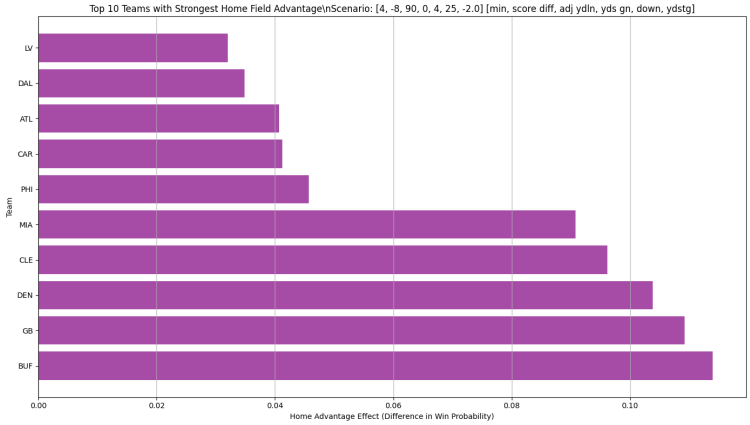



Figure 14: Top 10 teams by estimated home-field advantage effect (Scenario 2)

In the third scenario, the game context is highly favorable for the team in possession. This represents a near-certain win situation under typical circumstances. Figure 15 shows that all teams exceed a win probability of 0.95, with Philadelphia (PHI), Buffalo (BUF), and Kansas City (KC) reaching the upper limit of the scale. These results are consistent with expectations, as teams leading late in the game while in favorable field position generally exhibit very high likelihoods of winning.

Similarly, when the same teams are treated as the away team with possession, the win probabilities remain extremely high (Figure 16). While the magnitude of win probability is slightly lower on average compared to the home setting, top-performing teams still approach the 0.95 range. This indicates that even without home-field advantage, the scenario is favorable enough to yield nearly certain wins.

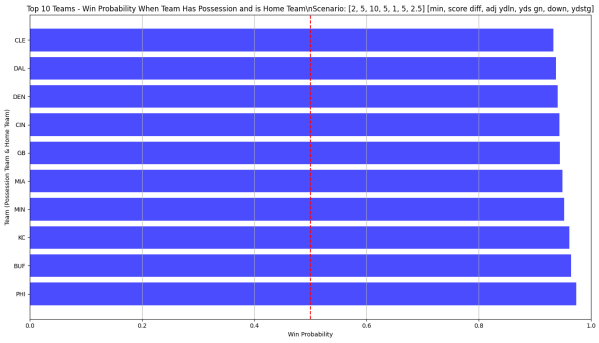



Figure 15: Top 10 teams by predicted win probability at home (Scenario 3)

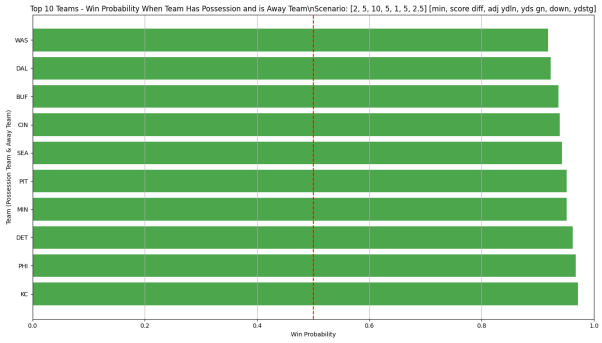



Figure 16: Top 10 teams by predicted win probability at away (Scenario 3)

It can be seen from Figure 17 that Carolina (CAR) and Cleveland (CLE) exhibit the most substantial increases in win probability from being the home team, suggesting they benefit the most from home-field conditions in high-leverage endgame situations. However, it can also be interpreted as Carolina has the highest chance of losing even in this scenario if they are playing away. While overall effects are smaller than in more uncertain scenarios, these variations still reflect meaningful boosts for selected teams.

Across all three scenarios, the models show sensitivity to game state, with win probability declining sharply under adverse conditions. However, team-specific home advantages are more pronounced in moderate and favorable contexts, while becoming negligible in extreme deficits.

## 8.4 Correlation with Fan Attendance

To further explore the sources of variation in home-field advantage across teams, one factor was chosen - fan attendance. The rationale is that a larger or more enthusiastic home crowd may provide psychological

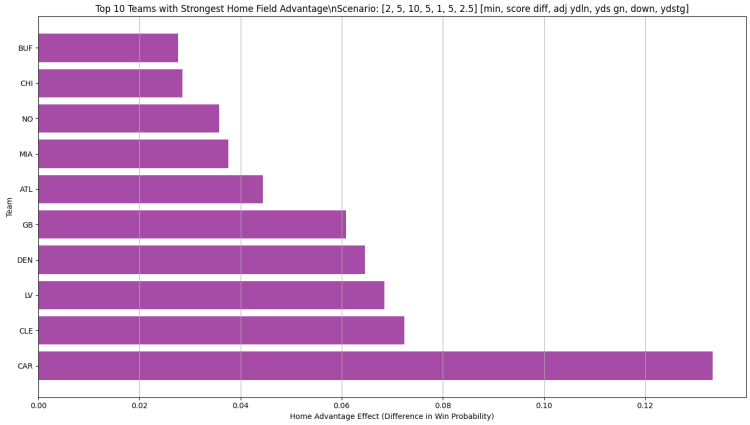



Figure 17: Top 10 teams by estimated home-field advantage effect (Scenario 3)

support for the home team and exert pressure on opponents, thereby amplifying the home-field effect.

To assess this relationship, each team's average home-field advantage was plotted against its three-year average home attendance ranking. A Kendall's tau correlation coefficient was computed to quantify the association.

According to Figure 18, the analysis resulted in a positive correlation coefficient of $\tau = 0.217$, with a p-value of 0.089. Although the p-value does not meet the conventional threshold of 0.05 for statistical significance, the positive trend still suggests a potential association between higher attendance and stronger home-field advantage.

This finding indicates that teams with larger home crowds may enjoy a slightly elevated performance edge, consistent with the idea that crowd size can influence game dynamics. However, the moderate strength of the correlation and its borderline significance imply that attendance is likely one of several contributing factors. Other plausible influences include travel distance for visiting teams, stadium altitude, climate conditions, and venue-specific familiarity or traditions.

Overall, while the evidence is not conclusive, the results point toward a multifaceted origin of home-field advantage, with fan attendance being one observable dimension worthy of further investigation.

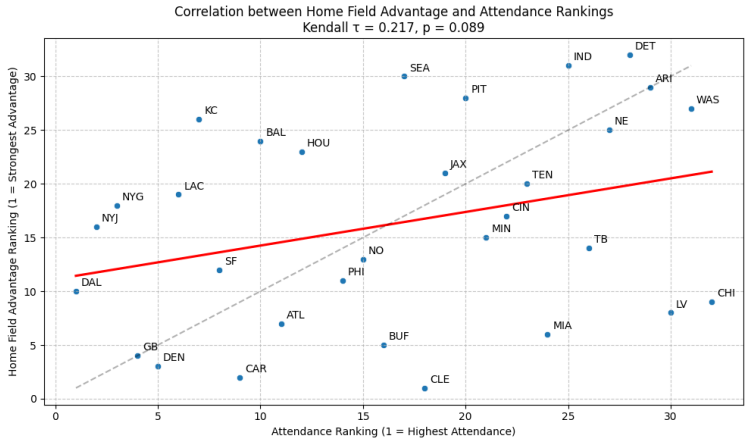



Figure 18: Correlation between Home Field Advantage and Fan Attendance Rankings

# 9 Discussion

## 9.1 Implications for Real-World Applications

The results from both models provide a flexible probabilistic framework for evaluating win probability in real time during NFL games. The hierarchical Bayesian structure allows the model to incorporate both team-specific strengths and game-state features, enabling dynamic adjustments as the game evolves. This framework can be particularly valuable in settings such as live sports analytics dashboards, in-game coaching decisions, and real-time betting markets.

The model's ability to disentangle team effects and home-field advantages also offers insights for strategic planning across seasons. For instance, identifying teams that consistently over- or underperform on the road can inform scheduling strategies or preseason adjustments. Additionally, the explicit quantification of uncertainty via posterior distributions supports more informed decision-making compared to point-estimate-based models.

## 9.2 Limitations & Assumptions

Despite its strengths, several limitations should be noted. First, the model assumes conditional independence between plays, disregarding temporal dependencies such as momentum, play-calling sequences, or clock management strategies. While this simplification facilitates tractable inference, it may understate the role of game flow in determining win probability.

Another limitation lies in the fixed structure of covariates. The model uses six continuous predictors—minutes remaining, score differential, field position, yards gained, down, and distance to first down—which may not capture the full strategic nuance of each play. Factors such as timeouts, penalty context, or formation types are excluded, potentially limiting predictive performance.

Additionally, while the varying home advantage model captures inter-team heterogeneity, it does not explicitly account for season-specific variations or team interactions. The hierarchical priors implicitly smooth these effects, but future models may benefit from a more granular temporal or interaction-based structure.

Furthermore, while the posterior distributions effectively capture parameter uncertainty, some confidence intervals, particularly for team-specific home advantage effects, remain relatively wide. This suggests that while the model captures broad patterns reliably, individual estimates can be noisy, reflecting limitations in sample size or feature coverage. Future work could incorporate additional covariates or longer time horizons to narrow posterior uncertainty and improve inference precision.

Lastly, as with all Bayesian hierarchical models, results are influenced by prior specifications. Although relatively uninformative priors were used, more robust sensitivity analyses could assess the stability of posterior estimates under alternative prior choices.

## 9.3 Future Work

Several extensions may further enhance the model's applicability and robustness. One direction is the integration of temporal dynamics using sequential models such as hidden Markov models, dynamic generalized linear models, or Bayesian state-space models. These approaches could capture evolving momentum or drive-based structures within games.

Incorporating additional covariates such as team possession history, player-level data, weather conditions, or betting market expectations—may also enrich the model's predictive capacity. Advances in real-time data acquisition and natural language processing can make these extensions more tractable.

Finally, the moderate correlation found between home-field advantage and fan attendance opens opportunities to explore broader sociological or logistical determinants of home performance—such as travel fatigue, stadium architecture, or fan engagement metrics—through mixed-effects or causal frameworks.

# 10 Conclusion

This study developed and evaluated Bayesian models for real-time win probability prediction in the NFL, leveraging hierarchical structures to capture both fixed game-state effects and latent team-specific character-

istics. Two models were considered: one with a fixed home-field advantage across teams and another allowing for varying home advantages. Both were trained on play-by-play data spanning three seasons (2022-2024), and posterior inference was conducted using modern probabilistic programming tools.

Model comparison revealed comparable predictive performance in terms of accuracy, AUC, Brier score, and log loss. However, the model with varying home-field advantage offered added interpretability and richer insights, particularly in identifying teams with stronger or weaker home effects. Posterior analysis of model parameters confirmed the dominance of score differential as the primary game-state predictor and highlighted meaningful heterogeneity in team strength and home advantage across franchises.

Scenario-based analyses further demonstrated how win probabilities adapt under varying game contexts, enabling targeted evaluation of team performance under pressure or with positional advantage. Additionally, an exploratory correlation analysis suggested a modest but positive relationship between home-field advantage and average fan attendance, pointing toward broader contextual factors influencing team outcomes.

By combining interpretability, flexibility, and uncertainty quantification, this modeling framework provides a strong foundation for real-time sports analytics. It contributes to the growing body of work at the intersection of Bayesian modeling and sports performance prediction and offers a viable path forward for deployment in practical, data-driven decision-making tools for teams, analysts, and fans.

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